Roll | | | | | | | | No.

EVEN END SEMESTER EXAMINATION- 2025

Name of the Course: Introduction to Unmanned Aerial System
Semester: IV
Name of thePaper: Introduction to Unmanned Aerial System *Paper Code: TAS 411*

Time: 3 Hours Maximum Marks: 100

Note:
(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	Explain in detail about the following: i. Waypoints ii. Geofencing iii. Airborne collision avoidance system iv. Ground control system v. Frist Person View	CO2
(b)	Discuss in detail about different types of electronic components used in Remotely Controlled aircraft.	
(c)	Compare the advantages and limitations of Li-Po and Li-ion batteries.	
Q2	(20 marks)	
(a)	Discuss the UAVs cybersecurity challenges in autonomous mission. How can AI threat detection mitigate risk?	
(b)	Explain the various measures of UAVs autonomy and discuss how these affect UAVs performance in unstructured environment?	CO3
(c)	What do you mean by adaptive flight control system? What are the advantages and limitations of this system?	
Q3	(20 marks)	
(a)	Assess the operational advantages and limitations of swarm intelligence in autonomous UAVs coordination and decision making.	CO4
(b)	Critically analyze the comparative performance of different UAVs communication system in terms of latency, reliability and operational range.	
(c)	Classify UAVs generated data products and critically evaluate their significance across different application domains.	

Q4	(20 marks)	CO 6
(a)	Explain the key features of Drone rule 2021 introduced by the government of India. How do these rules support the growth of UAVs industry in India?	
(b)	Analyze the interplay between ethical, legal and privacy issues in UAVs surveillance, citing real world policy examples.	
(c)	Discuss the role of Air Traffic control in managing mixed-traffic airspace involving manned and unmanned systems.	
Q5	(20 marks)	CO 5
(a)	Assess the global UAVs market trends and predict the future trajectory of UAVs in adoption in defense, commercial and civil sector.	
(b)	Conduct a SWOT analysis of UAVs technology in the context of its scalability, ethical dilemmas and regulatory bottle necks	
(c)	Evaluate the effectiveness of UAVs in post disaster response, highlighting their integration with geospatial intelligence systems.	